Lesson 27: Describing Single Variables

English Stage 4 (Psychology) · Unit 4: Statistics in Research Reports

Adapted from Research Methods in Psychology (4th American edition) by Rajiv S. Jhangiani, I-Chant A. Chiang, Carrie Cuttler and Dana C. Leighton (CC BY-NC-SA 4.0): 12.1 Describing Single Variables.

In this lesson

Before researchers test relationships, they describe each variable on its own: its distribution, its central tendency, its variability and the position of single scores. You will compute these statistics and practise the English for describing distributions and for reading formulas and numbers aloud.

Learning goals

After this lesson you should be able to:

- Display a distribution in a frequency table and describe its shape in words.
- Compute the mean, median and mode, and explain why the median suits highly skewed distributions.
- Compute the range, standard deviation, percentile ranks and *z* scores, and explain why software divides by $N - 1$.
- Read formulas aloud and report descriptive statistics in academic sentences.

Key language

Term or phrase	Meaning	Example
descriptive statistics	techniques for summarizing and displaying data	It is also important to describe each variable individually.
distribution	the way the scores on a variable are distributed across its levels	the distribution of "number of siblings"
frequency table / histogram	a table / a graph showing each value and how often it occurs	the Rosenberg self-esteem table
skewed	asymmetrical: positively skewed = long positive tail; negatively skewed = long negative tail	reaction times
outlier	an extreme score far removed from the rest	one clinically depressed person among happy peers
central tendency	the middle of a distribution (mean, median, mode)	the point around which the scores tend to cluster
mean (M)	the sum of the scores divided by the number of scores	The mean is 245 ms.

Term or phrase	Meaning	Example
median	the value with half the scores below it and half above	In 2 3 3 4 8 12 14, the median is 4.
mode	the most frequently occurring score	The mode is 22.
variability; range	how much scores vary around the middle; highest minus lowest score	The range is 24 – 15 = 9.
standard deviation (*SD*); variance (SD^2)	the square root of the mean of the squared differences between the scores and the mean; the variance is that mean	$SD^2 = 3.50$; $SD = 1.87$
percentile rank	the percentage of scores below a particular score	a percentile rank of 80
z score	the score minus the mean, divided by the standard deviation	(110 – 100) / 15 = +0.67
tend to cluster around	are mostly close to	The self-esteem scores tend to cluster around the values of 20 to 22.
… standard deviations above / below the mean	distance from the mean in *SD* units	85 is one standard deviation below the mean.

Rules and explanations

The ideas

Displaying a distribution. A **frequency table** lists values from highest (top) to lowest and usually stops at the highest and lowest scores observed (the Rosenberg table shows 24 to 15, though the scale runs from 0 to 30). With many different scores, a **grouped** table uses ranges of equal width, usually five to 15 of them. A **histogram** shows the same information (*x*-axis: values; *y*-axis: frequency), without gaps between bars for quantitative variables (the gap at 17 means nobody scored 17) and with small gaps for categorical ones.

Shape. A distribution can be **unimodal** or **bimodal**, and **symmetrical** (mirror-image halves) or **skewed**. A *negatively* skewed distribution has its peak toward the upper end and a long negative tail; a *positively* skewed one has its peak toward the lower end and a long positive tail. An **outlier** may be a truly extreme score or an error (a misunderstanding, an equipment malfunction).

Central tendency

Measure	How to find it	Best use
mean, $M = \Sigma X / N$	add the scores, divide by their number	most common; useful in inferential statistics
median	order the scores and take the middle one; with an even number, the value halfway between the two middle scores (2 3 3 4 8 12 14 15: 6)	highly skewed distributions

Measure	How to find it	Best use
mode	the most frequent score	the only measure for categorical variables

In a unimodal symmetrical distribution the three are close; in a skewed one the mean is pulled toward the longer tail. **Book example:** reaction times of 200, 250, 280 and 250 ms have a mean of 245 ms. Adding a score of 5,000 ms raises the mean to 5,980 / 5 = **1,196 ms** (the book prints 1,445 ms, a calculation error). The point still holds: this mean is greater than 80% of the scores, while the median (250 ms) represents the group well.

Variability. The **range** is misleading with outliers (exam scores of 90 to 100 have a range of 10; one score of 20 makes it 80). The **standard deviation** is roughly the average distance between the scores and the mean: $SD = \sqrt{[\Sigma(X - M)^2 / N]}$. Subtract the mean from each score, square each difference, take the mean of the squares (the **variance**), then the square root. Book example: 3, 5, 4, 2, 7, 6, 5, 8 (M = 5): sum of squares 28, $SD^2 = 28/8 = 3.50$, $SD = 1.87$. Dividing by N describes the sample; software divides by $N - 1$ because a sample's SD tends to be a little lower than the population's (here 28/7 = 4.00, $SD = 2.00$).

Where is one score? The **percentile rank** of 23 in the self-esteem table is 80, because 32 of the 40 scores are lower. A ***z* score** = $(X - M) / SD$: with IQ scores (M = 100, SD = 15), 110 gives +0.67 and 85 gives −1.00. Outliers are sometimes defined as z scores beyond ±3.00. **PSYC 110:** in Lecture 13 (Why Are People Different?), Paul Bloom explains that IQ tests are scored so that 100 is average and 68% of people score between 85 and 115, that is, between $z = -1.00$ and +1.00.

Language focus

1. Reading formulas and numbers aloud

Written	Spoken
$M = \Sigma X / N$	"*M* equals the sum of *X* divided by *N*" (Σ: "sigma")
$\sqrt{3.50} = 1.87$	"the square root of three point five oh equals one point eight seven"
$z = -1.00$	"*z* equals minus one point oh oh" (or "negative one")

2. Describing a distribution. The book joins a statistic to its meaning:

- "The self-esteem scores **tend to cluster around** the values of 20 to 22."
- "The distribution … is unimodal, **meaning** it has one distinct peak." (*meaning* + definition)
- "A score of 110 is 0.67 standard deviations (**approximately two thirds of** a standard deviation) **above the mean**."

> **Note for Arabic speakers** - English uses a decimal **point** (1.87, "one point eight seven") and a **comma** to group thousands (5,000). Never write 1,87.

Practice

Part 1: Reading questions

1. In a frequency table, in what order are the levels listed, and why does the Rosenberg table stop at 24 and 15?
2. When should you use a grouped frequency table, and what rules apply to its ranges?
3. Describe a negatively skewed distribution in your own words.
4. Why do researchers often prefer the median for reaction times?
5. Why can the range be misleading?
6. Why do most software packages divide by $N - 1$?
7. Find the percentile rank of a score of 22 in the self-esteem table.
8. Find the mean and median of 200, 250, 280, 250 and 5,000 ms. Does your mean agree with the book?

Part 2: Exercises from the book (chapter exercises, section 12.5)

1. Practice: Make a frequency table and histogram for the following data. Then write a short description of the shape of the distribution in words. 11, 8, 9, 12, 9, 10, 12, 13, 11, 13, 12, 6, 10, 17, 13, 11, 12, 12, 14, 14
2. Practice: For the data in Exercise 1, compute the mean, median, mode, standard deviation, and range.
3. Practice: Using the data in Exercises 1 and 2, find the percentile ranks for scores of 9 and 14; the z scores for scores of 8 and 12.

Part 3: Language exercise

- A. Write each as one academic sentence: (1) self-esteem mode = 22; (2) IQ score 130, $M = 100$, $SD = 15$; (3) self-esteem score 23, with 32 of 40 scores lower; (4) a distribution with its peak at the low end and a long tail toward high scores.
- B. Write in words how you would say: (a) $M = \Sigma X / N$; (b) $z = (85 - 100) / 15 = -1.00$.

Speak with your tutor

- "Let me explain why the mean misleads in a skewed distribution, using the reaction-time example. Correct my statistics and my English."
- "Give me eight scores. I will compute the standard deviation aloud, dividing by N and then by $N - 1$. Check my steps and how I say the numbers."
- "In PSYC 110 Lecture 13, Paul Bloom says that 68% of people score between 85 and 115 on IQ tests. Let me explain this with the mean, the standard deviation and z scores."

Writing task

Your research project: results, part 1 (describing each variable). In Lessons 27 to 32 you will write the results section of your sleep and memory study with this dataset (**invented data for**

practice). After learning 40 word pairs, 20 students were randomly assigned to a nap condition (a nap opportunity of up to 90 minutes; a sleep-tracking wristband recorded minutes asleep) or a wake condition (quiet rest for the same time). The dependent variable is the number of pairs recalled (out of 40).

Data (participants 1 to 10 in order)	
Nap: recall	24, 19, 22, 26, 21, 23, 18, 21, 25, 21
Nap: minutes asleep	70, 60, 80, 85, 55, 70, 75, 65, 70, 70
Wake: recall	18, 15, 21, 16, 12, 19, 16, 22, 14, 17

Write one paragraph (100 to 150 words) describing each variable: mean and standard deviation ($N - 1$, as software computes it), range, shape, and outliers (say how you defined them). Make a frequency table first. If you work on your own topic, invent a similar dataset (two groups of 10 and one quantitative variable), label it "invented data for practice" and ask your tutor to check every number; if your design is within-subjects or correlational, ask your tutor to adapt the data.

Checklist:

- Is the dataset labelled "invented data for practice"?
- Does every variable have a mean and a standard deviation, rounded to two decimal places?
- Did you give units (word pairs, minutes)?
- Did you describe each shape with this lesson's words (unimodal, symmetrical, skewed)?
- Did you say how you checked for outliers and what you found?
- Did you use the past tense for what happened (scores *ranged*, participants *slept*)?

Paste your draft into the chat and ask your tutor for feedback. Then rewrite it.

Watch online

- Rice Virtual Lab in Statistics, free analysis tools recommended by the book: https://onlinestatbook.com/stat_analysis/index.html
- VassarStats, free online calculators (the book's old address no longer works): http://vassarstats.net/

Answer key

Part 1

1. From highest to lowest. They usually stop at the highest and lowest scores in the data, and this data set only has scores from 15 to 24.
2. When there are many different scores across a wide range. The ranges must be of equal width, usually five to 15 of them.
3. (example answer) Most scores are high, so the peak is near the top of the range, and a few low scores form a long tail toward the low end.
4. Reaction times are often highly skewed: one very slow response (an inattentive participant) pulls the mean far from most scores but hardly affects the median.
5. One extreme score changes it greatly: the exam range goes from 10 to 80 although only one student differed.
6. A sample's *SD* tends to be a bit lower than the population's; dividing by $N - 1$ corrects this and better estimates the population *SD*.
7. 8 + 5 + 3 + 3 + 0 + 2 + 1 = 22 scores are lower; 22/40 = 55%, so the percentile rank is 55.
8. M = 5,980 / 5 = 1,196 ms; median = 250 ms. The book's 1,445 ms is wrong, but its conclusion stands: 1,196 is greater than 80% (4 of 5) of the scores.

Part 2

1. Frequency table (score: frequency): 17: 1; 16: 0; 15: 0; 14: 2; 13: 3; 12: 5; 11: 3; 10: 2; 9: 2; 8: 1; 7: 0; 6: 1. Histogram: tallest bar at 12; no bars at 7, 15 and 16. Shape (example answer): unimodal, with a peak at 12, and roughly symmetrical; most scores lie between 9 and 14, with single scores of 6 and 17 at the ends (17 is not an outlier by the ±3.00 rule: z = +2.36).
2. M = 229 / 20 = 11.45; median = 12; mode = 12; range = 17 – 6 = 11; *SD* = 2.36 dividing by N (2.42 dividing by $N - 1$).
3. Percentile ranks: 9 → 10 (2 of 20 scores lower); 14 → 85 (17 of 20 lower). z scores (*SD* = 2.36): 8 → (8 – 11.45) / 2.36 = –1.46; 12 → +0.23. (With *SD* = 2.42: –1.43 and +0.23.)

Part 3 (example answers)

- A1. The mode was 22: more students had this score than any other.
- A2. An IQ score of 130 is two standard deviations above the mean (z = +2.00).
- A3. Students who scored 23 scored at the 80th percentile.
- A4. The distribution is positively skewed, with its peak toward the lower end of its range and a relatively long positive tail.
- B. (a) "M equals the sum of X divided by N." (b) "z equals eighty-five minus one hundred, divided by fifteen, which equals minus one point oh oh."

Writing task (*SD* with $N - 1$; the N version in brackets)

Variable	*M*	*SD*	Median	Mode	Range
Recall, nap	22.00	2.54 [2.41]	21.50	21	8 (18 to 26)
Recall, wake	17.00	3.09 [2.93]	16.50	16	10 (12 to 22)
Minutes asleep	70.00	8.82 [8.37]	70.00	70	30 (55 to 85)

Example (no *z* score is beyond ±2): "In the nap condition, recall ranged from 18 to 26 word pairs (*M* = 22.00, *SD* = 2.54); in the wake condition, it ranged from 12 to 22 (*M* = 17.00, *SD* = 3.09). Both distributions were unimodal and roughly symmetrical, and no *z* score exceeded ±3.00. Participants in the nap condition slept a mean of 70.00 minutes (*SD* = 8.82)."

My notes